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CausalMixGPD: Bayesian Nonparametric Model with Optional Heavy-Tail Augmentaion and Causal Extensions

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Abstract

Heavy-tailed outcomes are commonly encountered in various scientific settings where rare extreme events dominate risk, cost, and policy recommendations. Although parametric models tend to adequately capture the center of a distribution, they often fail to characterize extreme outcomes, and non-parametric models lack principled extrapolation in tail regions. We develop `CausalMixGPD`, an R package for Bayesian modeling of heavy-tailed outcomes by integrating Dirichlet process mixtures for the center and generalized Pareto distributions for the tails in a unified framework. `CausalMixGPD` allows for both unconditional and covariate-dependent models, enabling flexible models of mixture weights and components as a function of predictors. Posterior computations were carried out using Markov chain Monte Carlo (MCMC) algorithms using the Nimble package in R. The framework is extended by allowing for separate modeling of outcome distributions for treatment groups. `CausalMixGPD` is designed for estimating and predicting causal quantities, such as average and quantile treatment effects, especially when quantiles are extreme. Moreover, this package provides a clustering framework for different kernel specifications. In R data analysis, it has demonstrated commendable performance in estimating extreme causal effects and clustering observations, thus providing a practical and extensible software framework for heavy-tailed outcome modeling.

Keywords: Bayesian nonparametrics; Dirichlet Process Mixture; Generalized Pareto Distribution; Extreme Value Theory; Cluster Analysis; Causal Inference; R; `nimble`.

1. Introduction

Causal inference is often summarized through average effects, but many applications require distributional comparisons between treatment regimes. In clinical trials, investigators may want to understand how a treatment changes outcomes across the outcome scale, including

effects among individuals in the upper tail where events are rare but consequences can be substantial. This shift from mean-based to distribution-based causal questions makes causal analysis tightly linked to how well we can estimate (and predict) outcome distributions under competing treatment assignments.

Dirichlet Process Mixtures (DPMs) provide a flexible way to model those outcome distributions because they accommodate multimodality and latent heterogeneity without fixing the number of mixture components in advance. They also induce random partitions through latent allocation variables, so density estimation and clustering are naturally linked. Bayesian cluster analysis is therefore typically summarized through label-invariant quantities such as pairwise co-clustering probabilities and representative partitions (Binder 1978; Dahl 2006). In regression settings, dependent Dirichlet process constructions allow mixture weights, component-specific parameters, or both to vary with predictors (MacEachern 1999, 2000; De Iorio, Müller, Rosner, and MacEachern 2004; Quintana, Müller, Jara, and MacEachern 2022), and predictor-dependent stick-breaking schemes provide a practical route to covariate-modulated cluster prevalence (Ren, Du, Carin, and Dunson 2011).

Heavy-tailed outcomes add a second challenge. The model must fit the center of the distribution well while still supporting principled upper tail extrapolation when data become sparse. CausalMixGPD addresses that combination by joining Bayesian nonparametric mixtures, optional generalized Pareto tails, clustering, and causal contrasts in a single workflow.

This article introduces an R package (CausalMixGPD) that connects Bayesian nonparametric modeling with EVT, clustering, and robust causal analyses in the presence of covariates (confounders). The package provides three simultaneous capabilities:

- **Bayesian nonparametric (BNP) density modeling and prediction with optional GPD tail:** CausalMixGPD fits DPMs using a user-chosen kernel below a certain threshold (bulk part), and can optionally splice a generalized Pareto tail above the threshold (tail part) to enable the estimation of tail quantities and posterior predictive inference for rare-event summaries.

- **Clustering under predictor dependent mixtures:** A dedicated clustering interface for both bulk-only and spliced models, supporting weight dependence, parameter dependence, or both, and returning label-invariant clustering summaries through posterior similarity matrices and Dahl representative partitions (Binder 1978; Dahl 2006).

- **Causal estimation and prediction for randomized controlled trials (RCTs) and observational studies:** The final goal of this package is to provide causal estimands by modeling treatment-specific potential outcome densities and reporting contrasts as functionals of the posterior over densities. Framework supports both randomized and observational settings, with the latter using a flexible Bayesian propensity score model to adjust for confounding.

Overall, CausalMixGPD offers a unified workflow for density estimation, prediction, and causal reporting when tail behavior is of significance. The package is implemented in R with computationally intensive components coded in C++ and integrated via the `nimble` framework for model specification and posterior simulation (R Package Authors 2025b). The remainder of this article is organized as follows: Section 2 provides background on the key modeling components, including DPMs, GPDs,and the spliced bulk-tail construction. Section 3 gives

an overview of the package's main functions and user interface. Section 4 and Section 5 illustrates the package's capabilities through applications to causal inference and clustering on R datasets, respectively. Finally, Section 6 concludes with a discussion of limitations and future directions.

2. Background and Preliminaries

Let the outcome–covariate pair be defined by

$$(Y, X) \in \mathbb{R} \times \mathbb{R}^p. \tag{1}$$

For a fixed covariate value x, the conditional cumulative distribution function (CDF) and density are

$$F(y \mid x) = \Pr\,(Y \leq y \mid X = x), \qquad f(y \mid x) = \frac{\partial}{\partial y} F(y \mid x), \tag{2}$$

and the conditional quantile function is defined by

$$Q(\tau \mid x) = \inf\Big\{y \in \mathbb{R} : \, F(y \mid x) \geq \tau\Big\}, \qquad 0 < \tau < 1. \tag{3}$$

CausalMixGPD treats conditional distribution of Y given X as the primary inferential object. The conditional density is modeled using a spliced (two part) construction that combines a Non–Parametric model for the center of the distribution where data is substantial and an extreme value distribution for the tail where data is scarce. All posterior summaries reported by the package are functionals of estimated conditional distributions' parameters, including conditional quantiles, densities, or samples.

The following subsections provide background on these two components and their integration into a spliced model.

2.1. Spliced DPM–GPD Conditional Distribution

CausalMixGPD adopts a spliced (bulk–tail) construction that retains the DPM fit below a high threshold and replaces the upper tail by a GPD component (Frigessi, Haug, and Rue 2002; Behrens, Lopes, and Gamerman 2004; Coles 2001). Let

$$p_u(x) = F_{\mathrm{DP}}(u(x) \mid x; \Theta) \tag{4}$$

denote the bulk probability mass below a threshold ($u(x)$) under the DPM fit. The spliced conditional CDF is given as

$$F(y \mid x; \Theta, \Phi) = \begin{cases} F_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ p_u(x; \Theta) + \{1 - p_u(x; \Theta)\}\, F_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{5}$$

where $F_{\mathrm{DP}}(\cdot)$ is the DPM CDF on the bulk region and $F_{\mathrm{GPD}}(\cdot)$ is the GPD CDF on the exceedance region (Section 7.1). It can be easily verified that the CDF (5) is continuous at $u(x)$ and differentiating (5) yields the spliced density fucntion given as

$$f(y \mid x; \Theta, \Phi) = \begin{cases} f_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ \{1 - p_u(x; \Theta)\}\, f_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{6}$$

where the tail density is a scaled GPD density, with scaling ensuring normalization of $f(\cdot \mid x; \Theta, \Phi)$. The following figure illustrates the spliced density construction, showing the bulk DPM density, the GPD tail density, and the resulting spliced density.

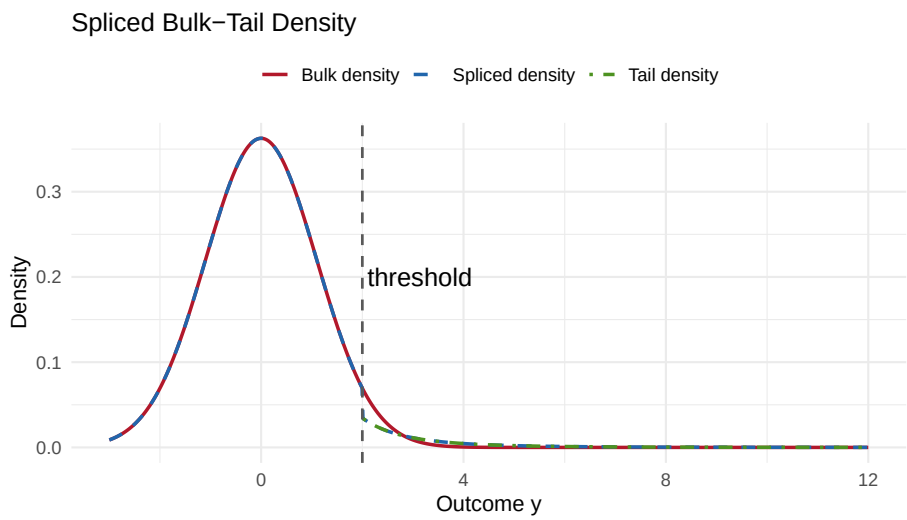

Figure 1: Illustration of the spliced bulk–tail density construction.

To find quantiles below the threshold i.e. for quantile index $\tau \in (0, p_u(x;\Theta)]$, one need to invert the CDF in (5) and obtain quantiles numerically. For $\tau \in (p_u(x;\Theta), 1)$, the index needs to be scaled in terms of threshold excedance probabilty which is defined as

$$\tilde{\tau}(x;\Theta) = \frac{\tau - p_u(x;\Theta)}{1 - p_u(x;\Theta)} \in (0,1), \tag{7}$$

and then the tail quantiles can be obtained by plugging in $\tilde{\tau}(x;\Theta)$ into the GPD quantile function in (50). Thus, the spliced quantile function is given by

$$Q(\tau \mid x;\Theta,\Phi) = \begin{cases} Q_{\mathrm{DP}}(\tau \mid x;\Theta), & \tau \le p_u(x;\Theta), \\ Q_{\mathrm{GPD}}(\tilde{\tau}(x;\Theta) \mid x;\Phi), & \tau > p_u(x;\Theta). \end{cases} \tag{8}$$

To make the spliced form and its associated functionals explicit, the parameters of the bulk and tail components are partitioned into two collections, denoted by Θ and Φ, respectively. Specifically, the bulk distribution defined in Equation (52) can be written as

$$f_{\mathrm{DP}}(y \mid x;\Theta) = \sum_{j=1}^{\infty} w_j \, k(y \mid x; \theta_j(x)), \tag{9}$$

with $\Theta = \{\beta_j\}_{j\ge1}$, where β_j denotes a set of component-specific regression coefficients. The corresponding kernel parameter at covariate value x is given by

$$\theta_j(x) = g(x^\top \beta_j),$$

where $g(\cdot)$ is a user-specified link function. The collection Θ may also include kernel parameters that do not depend on covariates. For example, under a Gaussian kernel, Θ may include

regression coefficients β_j^μ along with a component-specific variance parameter σ_j that does not depend on x.

For the tail component, define

$$\Phi = \{\beta_u, \beta_\sigma, \xi\}, \tag{10}$$

where the threshold and scale functions are specified as

$$u(x) = g_u(x^\top \beta_u), \qquad \sigma(x) = g_\sigma(x^\top \beta_\sigma),$$

with $g_u(\cdot)$ and $g_\sigma(\cdot)$ denoting user-specified link functions. The parameter ξ is the GPD shape parameter governing tail heaviness (see Section 7.1).

This package allows users to specify which parameters vary with covariates and which are treated as fixed or entirely determined by prior specifications (see Section 3).

2.2. Clustering under predictor-dependent Bayesian nonparametric mixtures

A common Bayesian nonparametric approach to conditional density modeling represents the collection $\{f(\,\cdot \mid x) : x \in \mathcal{X}\}$ through predictor-indexed random probability measures. Within the Dependent Dirichlet Process (DDP) framework of MacEachern (1999, 2000), one specifies a family $\{G_x : x \in \mathcal{X}\}$ of random mixing measures whose dependence across x induces flexible borrowing of information in $f(\,\cdot \mid x)$.

A generic mixture representation is

$$f(y \mid x) \;=\; \int k(y \mid x, \theta)\; dG_x(\theta), \tag{11}$$

where $k(\cdot \mid x, \theta)$ is a kernel and G_x is a predictor-indexed mixing measure. When G_x is discrete (as under stick-breaking constructions), latent parameters exhibit ties, inducing a random partition of observations. Introduce latent cluster-label variables z_i and component parameters $\{\theta_j\}_{j\geq 1}$ such that

$$y_i \mid z_i, \{\theta_j\}, x_i \sim k(\,\cdot \mid x_i, \theta_{z_i})\,. \tag{12}$$

Clustering is then characterized by the posterior distribution of the induced partition $\Pi = \{z_1, \ldots, z_n\}$ and summarized in a label-invariant manner using co-clustering functionals (Binder 1978; Dahl 2006).

In the clustering component considered here, we allow three forms of predictor dependence, each corresponding to a particular construction of G_x (equivalently, predictor dependence in the weights and/or atoms of a stick-breaking representation). In all cases, the conditional density admits the series form

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,, \tag{13}$$

with $w_j(x) \geq 0$ and $\sum_{j=1}^{\infty} w_j(x) = 1$ for all x.

- **Weight dependence only:** predictors modulate mixture weights while atoms are shared across x. Predictors may affect only the mixing weights, with atoms shared across x:

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid \theta_j)\,. \tag{14}$$

  This specification is appropriate when covariates modulate subpopulation prevalence while within-subpopulation distributions remain stable. A common approach uses predictor-dependent stick-breaking, where stick lengths are transformed regressions in x; the logistic stick-breaking process constructs these lengths via logistic regressions and maps them to weights (Ren *et al.* 2011). Such models can be viewed as practical DDP instances in which dependence is introduced through smoothly varying weights (MacEachern 2000; Quintana *et al.* 2022).

- **Atom dependence only:** predictors modulate component-specific kernel parameters while weights are shared across x Alternatively, predictors may enter through component-specific kernel parameters while the weights remain global:

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j\, k(y \mid x, \theta_j)\,. \tag{15}$$

  Here, θ_j typically indexes cluster-specific regression parameters (with appropriate link functions to enforce constraints), so that clusters correspond to distinct regression surfaces while w_j retains the interpretation of global prevalence. This structure aligns with DDP constructions that share weights while allowing predictor-indexed atoms or atom transformations (MacEachern 1999, 2000), and is closely related to analysis of variance (ANOVA)-type dependent priors for structured collections of random measures (De Iorio *et al.* 2004).

- **Weight and atom dependence:** predictors modulate both mixture weights and component-specific kernel parameters The most flexible specification allows predictors to influence both prevalence and within-cluster behavior:

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,. \tag{16}$$

  This corresponds to DDP-type models in which both selection probabilities and component-specific conditional structure vary with predictors (Quintana *et al.* 2022).

Posterior clustering summaries are reported using label-invariant functionals of Π. The posterior similarity matrix (PSM) has entries

$$S_{i\ell} \;=\; \Pr(z_i = z_\ell \mid \text{data}), \qquad i, \ell = 1, \ldots, n, \tag{17}$$

estimated by Monte Carlo averages over MCMC output. A representative partition may be obtained by selecting the sampled partition whose adjacency matrix is closest to the PSM under squared loss (Dahl 2006), yielding a stable point estimate while retaining the PSM as the primary uncertainty summary.

2.3. Causal Inference: Notation, Identification, and Reported Estimands

I use the potential outcomes framework (Neyman 1990; Rubin 1974; Imbens and Rubin 2015). Let A denote a binary treatment indicator, let X denote observed pre-treatment covariates, and let Y denote the observed outcome. Each unit has two potential outcomes, $Y(1)$ and $Y(0)$, and the observed outcome satisfies

$$Y = A\,Y(1) + (1 - A)\,Y(0). \tag{18}$$

The treatment-specific conditional distribution functions are the fundamental inferential objects:

$$F_a(y \mid x) = \Pr\{Y(a) \le y \mid X = x\}, \qquad a \in \{0, 1\}. \tag{19}$$

In this package, the corresponding conditional densities are

$$f_a(y \mid x) = \frac{\partial}{\partial y} F_a(y \mid x) \tag{20}$$

where $f_a(y \mid x)$ assumes the spliced DPM structure in (6). Conditional quantiles are defined by inversion of the conditional CDF:

$$Q_a(\tau \mid x) = \inf\{y : F_a(y \mid x) \ge \tau\}, \qquad \tau \in (0, 1). \tag{21}$$

Conditional means are functionals of the conditional density:

$$\mu_a(x) = \mathrm{E}\{Y(a) \mid X = x\} = \int y\, f_a(y \mid x)\, dy. \tag{22}$$

Marginal (standardized) distributions are obtained by averaging conditional distributions over a covariate distribution. The package reports population-standardized and treated-standardized functionals:

$$F_a^{\mathrm{m}}(y) = \mathrm{E}\{F_a(y \mid X)\}, \qquad F_a^{\mathrm{t}}(y) = \mathrm{E}\{F_a(y \mid X) \mid A = 1\}. \tag{23}$$

The corresponding marginal quantiles are

$$Q_a^{\mathrm{m}}(\tau) = \inf\{y : F_a^{\mathrm{m}}(y) \ge \tau\}, \qquad Q_a^{\mathrm{t}}(\tau) = \inf\{y : F_a^{\mathrm{t}}(y) \ge \tau\}. \tag{24}$$

Marginal means are

$$\mu_a^{\mathrm{m}} = \mathrm{E}\{\mu_a(X)\}, \qquad \mu_a^{\mathrm{t}} = \mathrm{E}\{\mu_a(X) \mid A = 1\}. \tag{25}$$

Identification of these quantities depends on the study design. In randomized experiments, randomization together with positivity implies exchangeability between treatment assignment and potential outcomes (Imbens and Rubin 2015). In observational studies, we rely on the following conditions (Rosenbaum and Rubin 1983; Imbens and Rubin 2015):

- **Stable Unit Treatment Value Assumption (SUTVA):** no interference between units and no hidden versions of treatment.
- **Consistency:** if $A = a$, then the observed outcome equals the corresponding potential outcome, $Y = Y(a)$.

- **Conditional ignorability:** treatment is as-if randomized given covariates,

$$(Y(0), Y(1)) \perp A \mid X. \tag{26}$$

- **Overlap (positivity):** each covariate profile has positive probability of receiving either treatment,

$$0 < \Pr(A = 1 \mid X = x) < 1 \tag{27}$$

  for covariate values of interest.

Under these conditions, the treatment-specific conditional distributions are identified from the observed data:

$$F_a(y \mid x) = \Pr(Y \leq y \mid A = a, X = x), \qquad a \in \{0, 1\}. \tag{28}$$

In observational settings, we use the propensity score (PS) to support the design stage (Rosenbaum and Rubin 1983). The PS is the treatment assignment probability given covariates:

$$\rho(x) = \Pr(A = 1 \mid X = x). \tag{29}$$

A key property of the true propensity score is that conditioning on it balances covariates across treatment groups:

$$A \perp X \mid \rho(X). \tag{30}$$

With estimated scores this balancing property is approximate in finite samples. Following the two-stage strategy described in the thesis, we first estimate $\rho(x)$ using only (A, X) and then augment the outcome model covariates with the estimated score (Rosenbaum and Rubin 1983):

$$\widehat{\rho}(x) \approx \rho(x), \qquad r(x) = \left(x^{\mathsf{T}}, \widehat{\rho}(x)\right)^{\mathsf{T}}. \tag{31}$$

The conditional outcome distributions are then modeled separately within each treatment arm as functions of the augmented predictor $r(x)$, and causal contrasts are computed as functionals of the estimated treatment-specific distributions.

The package reports the following causal estimands:

- **Average Treatment Effect (ATE):**

$$\mathrm{ATE} = \mu_1^{\mathrm{m}} - \mu_0^{\mathrm{m}}. \tag{32}$$

- **Average Treatment Effect on the Treated (ATT):**

$$\mathrm{ATT} = \mu_1^{\mathrm{t}} - \mu_0^{\mathrm{t}}. \tag{33}$$

- **Quantile Treatment Effect (QTE):** for any $\tau \in (0, 1)$ (Firpo 2007),

$$\mathrm{QTE}(\tau) = Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau). \tag{34}$$

- **Quantile Treatment Effect on the Treated (QTT):** for any $\tau \in (0, 1)$,

$$\mathrm{QTT}(\tau) = Q_1^{\mathrm{t}}(\tau) - Q_0^{\mathrm{t}}(\tau). \tag{35}$$

- **Conditional Average Treatment Effect (CATE):**

$$\mathrm{CATE}(x) = \mu_1(x) - \mu_0(x). \tag{36}$$

- **Conditional Quantile Treatment Effect (CQTE):** for any $\tau \in (0, 1)$,

$$\mathrm{CQTE}(\tau \mid x) = Q_1(\tau \mid x) - Q_0(\tau \mid x). \tag{37}$$

This section established the main probabilistic and causal building blocks used in the remainder of the work. I first defined conditional distributional targets and introduced a DPM model to flexibly estimate bulk conditional outcome densities. I then reviewed generalized Pareto modeling for threshold exceedances and constructed a spliced DPM–GPD conditional distribution that preserves the estimated bulk while enabling principled tail extrapolation. Finally, we introduced the potential outcomes framework and the treatment-effect estimands reported in the package, showing how marginal and conditional effects can be obtained as functionals of treatment-specific conditional distributions (with propensity-score augmentation supporting confounding adjustment in observational studies). Subsequent chapters build on these preliminaries to specify the full hierarchical model, posterior computation, and empirical evaluation of tail-aware distributional causal effects.

3. Package Overview: Workflow, Model, and Inference

This section describes the main workflow provided by CausalMixGPD for one-arm conditional modeling and prediction when a generalized Pareto tail is spliced above a covariate-dependent threshold. The causal extension reuses the same modeling, estimation, and prediction machinery after introducing treatment-specific outcome models and a propensity score component; we defer those additions to the causal section.

3.1. Augmented DPM–GPD model, likelihood, and priors

Let the observed data be

$$\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n,$$

```
data("nc_posX100_p3_k2", package = "CausalMixGPD")
dat <- data.frame(y = nc_posX100_p3_k2$y, nc_posX100_p3_k2$X)
```

with $x_i \in \mathbb{R}^p$ denoting the covariate vector (including an intercept if desired). The target conditional distribution $F(\cdot \mid x)$ and its spliced DPM–GPD construction are defined in Section 2, with the spliced CDF and density given in (5)–(6). Posterior computation in CausalMixGPD approximates the bulk DPM by a finite truncation with J mixture components,

$$f_{\mathrm{DP}}(y \mid x; \Theta) \;\approx\; \sum_{j=1}^{J} w_j \, k(y \mid x; \theta_j), \qquad j = 1, \ldots, J,$$

where j indexes mixture components, $w_j \geq 0$, and $\sum_{j=1}^J w_j = 1$. Kernel parameters can be constant or covariate-dependent through link specifications of the form

$$g(\psi_j(x)) = x^\top \beta_{j,\psi}, \tag{38}$$

where $g(\cdot)$ enforces parameter support (e.g., positive scale parameters).

```
mcmc_fixed
```

```
fit <- dpmgpd(
  formula = y ~ x1 + x2 + x3,
  data = dat,
  backend = "sb",
  kernel = "lognormal",
  components = 5,
  mcmc = mcmc_fixed
)
```

For the spliced model, let $u(x)$ be the threshold, $\sigma(x) > 0$ the tail scale, and $\xi \in \mathbb{R}$ the tail shape (as in Section 2.1). Define $\delta_i = \mathbf{1}\{y_i > u(x_i)\}$. Using the spliced density (6) with the truncated bulk mixture, the likelihood is

$$L(\Theta, \Phi; \mathcal{D}) = \prod_{i=1}^{n} f(y_i \mid x_i; \Theta, \Phi), \tag{39}$$

where Θ collects the truncation-level bulk parameters $\{w_j, \theta_j\}_{j=1}^{J}$, and Φ collects the tail parameters governing $u(x)$, $\sigma(x)$, and ξ. Equivalently, the log-likelihood can be written as

$$\begin{aligned} \log L(\Theta, \Phi; \mathcal{D}) \;=\; & \sum_{i=1}^{n} (1 - \delta_i) \log f_{\mathrm{DP}}(y_i \mid x_i; \Theta) \\ & + \sum_{i=1}^{n} \delta_i \Big[\log\{1 - F_{\mathrm{DP}}(u(x_i) \mid x_i; \Theta)\} + \log f_{\mathrm{GPD}}(y_i \mid x_i; \Phi) \Big], \end{aligned} \tag{40}$$

where f_{GPD} is given in (49) and $F_{\mathrm{DP}}(\cdot \mid x)$ is the bulk mixture CDF induced by the truncated mixture.

Priors follow the decomposition implied by (53) and (54). Under the SB backend, the truncation-level weights are generated through Beta stick breaks (Section 2, (53)), which are represented in nimble using the Beta distribution in model code (see `dbeta`). Under the CRP backend, the latent-cluster-label representation is implemented using the CRP distribution in nimble (see `dCRP`), matching the predictive representation in (54). Kernel parameters and tail parameters are assigned weakly informative priors, with regression coefficients $\beta_{j,\psi}$ arising through the link representation (38). The same link construction is used for covariate-dependent $u(x)$ and $\sigma(x)$ when those are modeled as functions of x.

3.2. MCMC algorithm, convergence diagnostics, and parameter extraction

Posterior simulation targets $p(\Theta, \Phi \mid \mathcal{D}) \propto L(\Theta, \Phi; \mathcal{D})\, p(\Theta, \Phi)$ using the likelihood (39) and the prior structure implied by the SB or CRP representation (Section 2, (53)–(54)). Given a model specification, CausalMixGPD runs MCMC through nimble (de Valpine, Turek, Paciorek, Anderson-Bergman, Temple Lang, and Bodik 2017; R Package Authors 2025b) under the chosen backend; SB uses Beta stick breaks in model code (see `dbeta`) and CRP uses the CRP distribution for latent cluster labels (see `dCRP`). Users control the number of iterations, burn-in, thinning, and number of chains through the `mcmc` argument in the wrappers.

Convergence checking is supported through standard multi-chain diagnostics and trace-based visualizations. The model object supports `summary()` for posterior summaries, `plot()` for

diagnostic plots (trace, running mean, autocorrelation, and multi-chain diagnostics such as $\widehat{R}$ when multiple chains are used), and `params()` as a lightweight extractor returning posterior mean parameters reshaped to natural dimensions. Interval summaries used by `predict()` can be chosen as equal-tailed credible intervals or HPD intervals as described in Section 3.3 (Gelman, Carlin, Stern, Dunson, Vehtari, and Rubin 2013; Plummer, Best, Cowles, and Vines 2024). Diagnostic plotting relies on established MCMC diagnostic tooling (R Package Authors 2025a).

```
print(fit)
p_hat <- params(fit)
p_hat
```

Bulk-only special case (no GPD tail)

Setting `GPD=FALSE` removes the splicing step so that $f(y \mid x; \Theta) \equiv f_{\mathrm{DP}}(y \mid x; \Theta)$, and $L(\Theta, \Phi; \mathcal{D})$ reduces to the bulk-mixture likelihood in Θ. All estimation and prediction targets described below remain available, with tail-specific terms omitted. The wrapper `dpmix()` fits this bulk-only model, while `dpmgpd()` fits the augmented (spliced) model.

3.3. Posterior functionals, interval summaries and prediction

The estimands reported by CausalMixGPD are functionals of the estimated conditional distribution $F(\cdot \mid x)$ from Section 2. In this article we focus on four conditional functionals: density $f(y \mid x)$, survival $S(y \mid x) = 1 - F(y \mid x)$, quantiles $Q(\tau \mid x)$, and mean-type summaries.

Let $(\Theta^{(s)}, \Phi^{(s)})_{s=1}^{S}$ denote post-burn-in draws from the posterior. For density, survival, and quantiles, CausalMixGPD evaluates the corresponding draw-wise functional using the spliced formulas (5)–(8) (with the truncated bulk mixture), and then summarizes across draws. Specifically, for a generic scalar functional $T(\Theta, \Phi; x)$ (e.g., $f(y_0 \mid x)$ or $Q(\tau \mid x)$),

$$\widehat{T}(x) = \frac{1}{S} \sum_{s=1}^{S} T(\Theta^{(s)}, \Phi^{(s)}; x),$$

and uncertainty is summarized either by equal-tailed credible intervals (posterior quantiles) or by highest posterior density (HPD) intervals (shortest intervals covering the target posterior mass). Equal-tailed intervals and general Bayesian posterior summaries follow standard practice (Gelman *et al.* 2013); HPD intervals are computed from the MCMC draws using established implementations (Plummer *et al.* 2024).

Mean-type summaries are handled differently in the package. For `type="mean"` and `type="rmean"`, CausalMixGPD uses posterior predictive Monte Carlo within each posterior draw rather than direct plug-in evaluation. For a fixed covariate value x, within draw s the package generates

$$y_1^{(s)}, \ldots, y_M^{(s)} \;\sim\; f(\cdot \mid x; \Theta^{(s)}, \Phi^{(s)}),$$

and computes

$$\mu^{(s)}(x) = \frac{1}{M} \sum_{m=1}^{M} y_m^{(s)}, \qquad \mu_c^{(s)}(x) = \frac{1}{M} \sum_{m=1}^{M} \min\{y_m^{(s)}, c\},$$

corresponding to the conditional mean and the restricted mean at cutoff c, respectively. Posterior summaries and intervals are then formed from $\{\mu^{(s)}(x)\}_{s=1}^{S}$ or $\{\mu_c^{(s)}(x)\}_{s=1}^{S}$. Under the spliced model, the conditional mean is infinite when $\xi \geq 1$; accordingly, CausalMixGPD flags `type="mean"` as infinite when there is posterior mass with $\xi \geq 1$, while `type="rmean"` remains finite by construction.

Using the same fitted model `fit` from Section 3.1, we now compute posterior summaries on the original training covariates (default behavior of `predict()` when `newdata` is not supplied).

```
q_grid <- c(0.50, 0.90, 0.95, 0.99)
x_eval <- dat[1:20, c("x1", "x2", "x3")]

est_quant <- predict(fit, newdata = x_eval, type = "quantile", index = q_grid,
                     interval = "hpd", level = 0.95)
```

For a new covariate value $x^\star$, prediction is based on the posterior predictive distribution

$$p(y^\star \mid x^\star, \mathcal{D}) = \int f(y^\star \mid x^\star; \Theta, \Phi)\, p(\Theta, \Phi \mid \mathcal{D})\, d\Theta\, d\Phi.$$

The `predict()` method returns posterior summaries for the same functional targets as in Section 3.3. For `type="density"` and `type="survival"`, the user supplies a grid of y values and the method returns pointwise posterior means with interval bounds computed draw-by-draw. For `type="quantile"`, the user supplies quantile indices τ and the method returns posterior summaries of $Q(\tau \mid x^\star)$ computed using the spliced quantile representation in (8). For `type="mean"` and `type="rmean"`, prediction uses posterior predictive Monte Carlo as described above, with x replaced by $x^\star$ and with user-controlled M through `nsim_mean`.

```
q_levels <- c(0.25, 0.50, 0.75)
x_new <- as.data.frame(lapply(
  dat[, c("x1", "x2", "x3")],
  quantile,
  probs = q_levels,
  na.rm = TRUE
))
rownames(x_new) <- c("q25", "q50", "q75")
y_grid <- seq(0, 10, length.out = 200)
p_grid <- c(0.50, 0.90, 0.95, 0.99)
```

```
pdens <- predict(fit, newdata = x_new, y = y_grid, type = "density",
                 interval = "credible", level = 0.95)
```

```
psurv <- predict(fit, newdata = x_new, y = y_grid, type = "survival",
                 interval = "credible", level = 0.95)
```

```
pquant <- predict(fit, newdata = x_new, type = "quantile", index = p_grid,
                  interval = "hpd", level = 0.95)
```

3.4. Clustering extension

The mixture modeling framework introduced earlier induces a latent partition of the observations through the latent cluster labels z_i. While the primary modeling workflow of the package focuses on estimating conditional densities and causal functionals, the clustering extension instead treats this latent partition as the main object of interest. The package therefore provides a dedicated clustering interface that fits the same mixture models but returns clustering summaries derived from the posterior distribution of the latent cluster labels.

The clustering workflow mirrors the design used throughout the package. Users specify the response variable, optional covariates, the mixture kernel, and whether the model should use the bulk-only or the spliced likelihood. The resulting model is then fitted using the same MCMC framework used for the conditional and causal models. Consequently, clustering models share the same prior specification, kernel choices, and computational infrastructure as the rest of the package.

Two clustering wrappers are provided. The first constructs mixture models using only the bulk kernel distribution. The second uses the spliced bulk–tail likelihood introduced earlier, allowing the clustering structure to account for extreme observations through the generalized Pareto tail. Both interfaces accept the same model specification arguments as the main modeling functions, ensuring that clustering models can be constructed using the same syntax and options.

Three forms of covariate dependence are supported in clustering models. First, covariates may influence the mixture weights while the component distributions remain invariant across observations. In this configuration, clusters represent subpopulations whose prevalence varies with predictors. Second, the mixture weights may remain global while the component parameters depend on covariates through the same link structures used in the conditional density models. In this case clusters represent groups with distinct covariate-response relationships. Third, both the mixture weights and the component parameters may depend on covariates simultaneously, allowing clusters to capture both changing prevalence and covariate-dependent distributional behavior.

Posterior inference for clustering models uses the same sampling procedure as the main mixture models, jointly updating the component parameters and the latent cluster labels. Let $z^{(s)} = (z_1^{(s)}, \ldots, z_n^{(s)})$, $s = 1, \ldots, S$, denote the retained MCMC draws of the latent cluster-label vector after burn-in and thinning. Because mixture components are only identified up to label permutation, `CausalMixGPD` reports label-invariant clustering summaries based on pairwise co-clustering probabilities and a representative partition (Binder 1978; Dahl 2006).

The primary uncertainty summary is the posterior similarity matrix (PSM), computed entrywise as

$$\widehat{S}_{i\ell} = \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\left\{ z_i^{(s)} = z_\ell^{(s)} \right\}, \qquad i, \ell = 1, \ldots, n.$$

Thus, $\widehat{S}_{i\ell}$ is the posterior probability that observations i and ℓ belong to the same cluster.

A single representative partition is then extracted by selecting the sampled partition whose adjacency matrix is closest to the PSM under squared loss (Dahl 2006):

$$s^\star = \arg\min_{1 \le s \le S} \sum_{i=1}^{n} \sum_{\ell=1}^{n} \left(\mathbf{1}\left\{ z_i^{(s)} = z_\ell^{(s)} \right\} - \widehat{S}_{i\ell} \right)^2.$$

The representative labels are $\hat{z}_i = z_i^{(s^\star)}$, relabeled to consecutive integers $1, \ldots, K$, and the

reported cluster sizes are $n_k = \sum_{i=1}^{n} \mathbf{1}\{\hat{z}_i = k\}$.

When `return_scores = TRUE`, the package also returns a training membership score matrix. If $C_k = \{\ell : \hat{z}_\ell = k\}$ denotes cluster k in the representative partition, the unnormalized score for observation i and cluster k is

$$q_{ik} = \frac{1}{|C_k|} \sum_{\ell \in C_k} \widehat{S}_{i\ell},$$

and the normalized training score is

$$p_{ik} = \frac{q_{ik}}{\sum_{h=1}^{K} q_{ih}}.$$

The hard representative label is $\arg\max_{1 \le k \le K} p_{ik}$, while a simple cluster-certainty summary is $\max_{1 \le k \le K} p_{ik}$.

For a new observation $(y_i^{\text{new}}, x_i^{\text{new}})$, the package first forms within-draw component scores

$$r_{ij}^{(s)} \propto \omega_j^{(s)}(x_i^{\text{new}})\, f_j^{(s)}(y_i^{\text{new}} \mid x_i^{\text{new}}), \qquad j = 1, \ldots, J,$$

where $\omega_j^{(s)}(x)$ is the draw-specific gating weight when that quantity is available; otherwise the implementation falls back to sampled global weights, when present, or to the empirical component frequencies in draw s. For occupied components, these scores are mapped to representative clusters through

$$m_{jc}^{(s)} = \frac{1}{n_j^{(s)}} \sum_{\ell:\, z_\ell^{(s)} = j} \mathbf{1}\{\hat{z}_\ell = c\}, \qquad c = 1, \ldots, K,$$

and then averaged over posterior draws:

$$p_{ic}^{\text{new}} = \frac{1}{S} \sum_{s=1}^{S} \sum_{j=1}^{J} r_{ij}^{(s)} m_{jc}^{(s)}.$$

Empty sampled components are assigned a uniform map over the K representative clusters. These are the quantities returned internally by `predict.dpmixgpd_cluster_fit()`. After row normalization, the predicted cluster label is $\arg\max_{1 \le c \le K} p_{ic}^{\text{new}}$.

The clustering workflow is illustrated below using one of the example data sets included with the package.

```r
data("nc_realX100_p3_k2", package = "CausalMixGPD")

dat_cl <- data.frame(y = nc_realX100_p3_k2$y, nc_realX100_p3_k2$X)

fit_cluster <- dpmix.cluster(
  y ~ x1 + x2 + x3,
  data       = dat_cl,
  kernel     = "normal",
  type       = "param",
  components = 8,
  mcmc = mcmc_fixed
)
```

Posterior clustering summaries can be obtained directly from the fitted object. The following code returns the PSM, the representative training partition, the training cluster sizes, and the associated certainty scores.

```
cluster_psm   <- predict(fit_cluster, type = "psm")
cluster_train <- predict(fit_cluster, type = "label", return_scores = TRUE)

cluster_sizes <- data.frame(
  cluster = names(summary(cluster_train)$cluster_sizes),
  n = as.integer(summary(cluster_train)$cluster_sizes)
)
cluster_certainty <- apply(cluster_train$scores, 1, max)
```

Cluster assignments for new observations are obtained from the posterior predictive score matrix in the same interface.

```
cluster_new <- predict(
  fit_cluster,
  newdata = dat_cl,
  type    = "label",
  return_scores = TRUE
)

cluster_new_certainty <- apply(cluster_new$scores, 1, max)
```

The same workflow applies to the spliced mixture model, which incorporates the generalized Pareto tail component when clustering data with heavy-tailed behavior.

```
fit_cluster_spliced <- dpmgpd.cluster(
  y ~ x1 + x2 + x3,
  data       = dat_cl,
  kernel     = "normal",
  type       = "both",
  components = 8,
  mcmc = mcmc_fixed
)
```

3.5. Causal extension and propensity score augmentation

This package extends the one-arm conditional mixture regression framework to two-arm causal studies by fitting an outcome model under treatment and under control, optionally augmenting the outcome covariates with a PS summary. Throughout this section we write the model in its default *augmented spliced* form (bulk mixture + GPD tail), using the same conditional distributional building blocks introduced earlier (so we do not restate the spliced density; see (6) and the corresponding conditional functionals defined in the previous sections).

Data, model components, and likelihood factorization

Let

$$\mathcal{D} = \{(y_i, a_i, \boldsymbol{x}_i)\}_{i=1}^n, \qquad a_i \in \{0, 1\},$$

where $a_i = 1$ denotes treatment and $a_i = 0$ denotes control. The causal interface consists of two pieces.

First, when PS augmentation is enabled, a binary regression model is used for the treatment mechanism

$$a_i \mid \boldsymbol{x}_i, \boldsymbol{\gamma} \sim \text{Bernoulli}(e_{\boldsymbol{\gamma}}(\boldsymbol{x}_i)), \tag{41}$$

$$h(e_{\boldsymbol{\gamma}}(\boldsymbol{x}_i)) = \gamma_0 + \boldsymbol{x}_i^\top \boldsymbol{\gamma}, \qquad h \in \{\text{logit}, \text{probit}\}, \tag{42}$$

which is consistent with the definition and balancing property in (29)–(30). The package also supports a `"naive"` option, in which $e(\boldsymbol{x})$ is taken to be constant (no covariate dependence).

Second, the outcome model is fit separately within each arm. Denote by $F_a(\cdot \mid \boldsymbol{r})$ the fitted conditional outcome distribution under arm $a \in \{0, 1\}$, where $\boldsymbol{r}$ is the covariate vector used by the outcome model. With PS augmentation, the package constructs an *augmented* covariate vector

$$\boldsymbol{r}(\boldsymbol{x}) = \Big(1,\ \boldsymbol{x}^\top,\ \psi(\widehat{e}(\boldsymbol{x}))\Big)^\top, \tag{43}$$

where $\widehat{e}(\boldsymbol{x})$ is a posterior summary of the PS (posterior mean or median, controlled by `ps_summary`), and $\psi(\cdot)$ is either $\psi(p) = \text{logit}(p)$ or $\psi(p) = p$ (controlled by `ps_scale`). When PS augmentation is disabled, $\boldsymbol{r}(\boldsymbol{x}) = (1, \boldsymbol{x}^\top)^\top$.

Within each arm, the conditional density $f_a(y \mid \boldsymbol{r})$ follows the same mixture regression + optional GPD tail specification described in the one-arm section (kernel choice, link structure $g(\cdot)$ for covariate-dependent parameters, and either `"sb"` or `"crp"` backend for the DPM). Let $\boldsymbol{\Theta}_a$ collect the arm-specific mixture and tail parameters. The arm-wise likelihood contributions are

$$L_a(\boldsymbol{\Theta}_a; \mathcal{D}) = \prod_{i:\, a_i = a} f_a(y_i \mid \boldsymbol{r}(\boldsymbol{x}_i); \boldsymbol{\Theta}_a)\,, \qquad a \in \{0, 1\},$$

with $f_a(\cdot \mid \cdot)$ given by the spliced construction already defined (bulk mixture below threshold and GPD exceedances above threshold). Practically, the implementation uses a two-stage workflow: (i) fit the PS model using (41)–(42), then (ii) compute $\widehat{e}(\boldsymbol{x}_i)$ and build $\boldsymbol{r}(\boldsymbol{x}_i)$ in (43) before fitting the two arm-specific outcome models.

Prior specification (PS then outcome)

For the PS regression coefficients, the default prior is independent Normal,

$$\gamma_0 \sim \mathcal{N}(m_0, s_0^2), \qquad \gamma_k \sim \mathcal{N}(m_0, s_0^2),$$

with defaults controlled by `ps_prior`.

For each arm's outcome model, priors follow the same structure as the one-arm case: a DPM prior (with either truncated stick-breaking or a CRP latent-cluster-label representation, implemented in `nimble` (de Valpine *et al.* 2017; R Package Authors 2025b)) and kernel-/tail-specific priors on the bulk and GPD parameters. Since the causal fit is simply a pair of one-arm fits (treated and control), all customization options for kernels, links, and tail inclusion apply arm-wise. The corresponding function arguments and printed summaries provide the quickest lookup for these controls and their code-level defaults.

MCMC algorithm, diagnostics, and parameter extraction

The causal fit runs MCMC in two stages, each with its own iteration controls.

(i) PS stage: posterior simulation for $\boldsymbol{\gamma}$ in (41)–(42) using `mcmc_ps` settings. The PS posterior draws are summarized to obtain $\widehat{e}(\boldsymbol{x}_i)$, which is optionally transformed via `ps_scale` and appended to the outcome covariates as in (43).

(ii) Outcome stage: arm-specific posterior simulation for $\boldsymbol{\Theta}_0$ and $\boldsymbol{\Theta}_1$ using `mcmc_outcome` settings, with either `"sb"` or `"crp"` backend.

For routine convergence checks, users can inspect trace plots and autocorrelation for monitored parameters and (when running multiple chains) compare between-chain and within-chain mixing. The package stores MCMC samples in the fitted object, and standard diagnostics can be applied after converting samples to `coda` objects (e.g., effective sample size, Geweke diagnostics, and HPD intervals via `coda::HPDinterval`; see the `coda` reference at the end of this message). Posterior parameter summaries can be extracted via the exported `params()` method, which returns arm-specific parameter tables for causal fits.

Estimation and interval summaries for causal contrasts

All reported causal quantities are computed as functionals of the fitted arm-specific conditional distributions. Let $T_a(\boldsymbol{x})$ denote any conditional functional under arm a (density, survival, quantile, or mean/restricted mean), computed from $F_a(\cdot \mid \boldsymbol{r}(\boldsymbol{x}))$ using the same definitions as in the one-arm case. The corresponding conditional causal contrast is

$$\Delta_T(\boldsymbol{x}) = T_1(\boldsymbol{x}) - T_0(\boldsymbol{x}).$$

The package computes $\Delta_T(\boldsymbol{x})$ *draw-by-draw* from posterior samples, then summarizes $\Delta_T(\boldsymbol{x})$ by a posterior mean/median and uncertainty intervals. Equal-tailed credible intervals are formed by posterior quantiles, while HPD intervals are computed using `coda::HPDinterval`.

The main causal estimators provided by the package are:

(a) Conditional effects at covariate profiles. For a user-supplied set of covariate rows $\{\boldsymbol{x}_m^\star\}_{m=1}^M$,

$$\mathrm{CATE}(\boldsymbol{x}_m^\star) = \mu_1(\boldsymbol{x}_m^\star) - \mu_0(\boldsymbol{x}_m^\star), \qquad \mathrm{CQTE}(\tau \mid \boldsymbol{x}_m^\star) = Q_1(\tau \mid \boldsymbol{x}_m^\star) - Q_0(\tau \mid \boldsymbol{x}_m^\star),$$

where $\mu_a(\boldsymbol{x}) = \mathbb{E}(Y \mid A = a, \boldsymbol{X} = \boldsymbol{x})$ and $Q_a(\tau \mid \boldsymbol{x})$ is the conditional quantile function. In heavy-tailed settings, $\mu_a(\boldsymbol{x})$ can be unstable when the implied tail shape yields infinite mean; the package therefore supports both `type="mean"` and `type="rmean"` for (restricted) mean-based effects, with restricted means computed as $\mathbb{E}[\min\{Y, c\} \mid A = a, \boldsymbol{X} = \boldsymbol{x}]$ at user-specified cutoff c.

(b) Marginal (standardized) summaries. The package reports marginal ATE and marginal QTE-style summaries by empirical standardization over covariate profiles:

$$\mathrm{ATE} = \frac{1}{n}\sum_{i=1}^{n}\Big\{\mu_1(\boldsymbol{x}_i) - \mu_0(\boldsymbol{x}_i)\Big\}, \tag{44}$$

$$\mathrm{QTE}(\tau) = Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau), \qquad Q_a^{\mathrm{m}}(\tau) = \inf\{y : F_a^{\mathrm{m}}(y) \geq \tau\}, \quad F_a^{\mathrm{m}}(y) = \frac{1}{n}\sum_{i=1}^{n} F_a(y \mid \boldsymbol{x}_i). \tag{45}$$

Treated-standardized analogues (ATT, QTT) are obtained by replacing the full-sample covariate empirical distribution with the treated-subsample empirical distribution $\{i : a_i = 1\}$.

In the current implementation, `ate()` and `att()` aggregate draw-wise conditional mean contrasts across rows, while `qte()` and `qtt()` compute draw-wise arm-specific marginal quantiles and then contrast them.

Posterior prediction at new covariate profiles

Prediction in the causal setup proceeds by producing arm-specific predictive summaries at new covariates and then differencing when required. For new covariates $\boldsymbol{x}^\star$, the package first predicts the PS (if enabled), forms $\boldsymbol{r}(\boldsymbol{x}^\star)$ via (43), and then evaluates the desired distributional functional under both arms. For `type = "mean"` and `type = "quantile"`, `predict()` returns treated-minus-control summaries and stores arm-specific predictions as attributes; for `type = "density"`, `"survival"`, and `"prob"`, it returns arm-specific summaries directly.

A recommended way to build an interpretable prediction grid is to use coordinate-wise empirical quartiles. Let $q_{k,p}$ be the pth sample quantile of covariate x_k for $p \in \{0.25, 0.5, 0.75\}$. The grid

$$\mathcal{X}_\star = \prod_{k=1}^{p} \{q_{k,0.25}, q_{k,0.5}, q_{k,0.75}\}$$

yields 3^p representative covariate profiles at which CATE, CQTE, survival contrasts, and other predictive contrasts can be reported.

User-facing workflow (code)

The following example uses the formula interface and the high-level wrapper, fitting the augmented spliced model by default.

```
data("causal_pos500_p3_k2", package = "CausalMixGPD")
dat <- causal_pos500_p3_k2
df  <- data.frame(y = dat$y, A = dat$A, dat$X)
```

```
cfit <- dpmgpd.causal(
  formula = y ~ x1 + x2 + x3,
  data = df,
  treat = "A",
  backend = "sb",
  kernel = "lognormal",
  components = 5,
  PS = "logit",
  ps_scale = "logit",
  ps_summary = "mean",
  mcmc_outcome = mcmc_fixed,
  mcmc_ps = mcmc_fixed
)
```

The fitted model object `cfit` contains the MCMC samples for both the PS and the arm-specific outcome models, and can be used to compute causal estimands and predictive summaries.

To estimate the ATE and QTE at the empirical covariate distribution, use the `ate()` and `qte()` functions, which compute the standardized contrasts in (44) and (45).

```
ate_hat <- ate(cfit, level = 0.90, interval = "credible")
qte_hat <- qte(cfit, probs = c(0.50, 0.90, 0.95), level = 0.90, interval = "hpd")
```

For subpopulation effects, the `cate()` and `cqte()` functions compute conditional contrasts at user-specified covariate rows.

```
qs    <- c(0.25, 0.50, 0.75)
Xgrid <- expand.grid(lapply(df[c("x1","x2","x3")], quantile, probs = qs))
```

The following code then produces conditional quantile and survival predictions at the grid points, returning treated-minus-control contrasts for quantiles and arm-specific summaries for survival.

```
pred_q <- predict(cfit, newdata = Xgrid, type = "quantile", p = c(0.50, 0.90))
pred_s <- predict(cfit, newdata = Xgrid, type = "survival", y = rep(4, nrow(Xgrid)))
```

Bulk-only causal mixtures (no GPD tail)

To run the same two-arm regression without a GPD tail, use the bulk-only wrapper `dpmix.causal()` (or set `GPD=FALSE` at bundle construction). All causal estimand and prediction functions (`ate`, `qte`, `cate`, `cqte`, and `predict`) operate the same way, now interpreting $F_a(\cdot \mid \boldsymbol{x})$ as the bulk mixture distribution.

4. Data analysis I: Clustering

This application demonstrates the clustering wing for a conditional mixture model under a bulk-only likelihood (no GPD tail). The objective is to infer latent groups of observational units implied by the fitted mixture model, and to summarize posterior clustering uncertainty in a label-invariant way. Throughout, clustering is reported via (i) the posterior similarity matrix (PSM) and (ii) a single representative partition derived from the posterior draws, together with out-of-sample cluster label prediction for new units.

4.1. Dataset and objective

We use the `Boston` housing data (MASS), treating each census tract as an observational unit. Let

$$\mathcal{D} = \{(y_i, \boldsymbol{x}_i)\}_{i=1}^n,$$

where y_i is the per-capita crime rate (`crim`) for tract i, and $\boldsymbol{x}_i$ is a vector of tract characteristics. Our aim is twofold: (i) to cluster tracts using a flexible conditional mixture model that can accommodate skewness and heterogeneity, and (ii) to assign cluster labels to new tracts using posterior predictive co-clustering evidence.

4.2. Preprocessing and train/test split

We split the data into a training set used to fit the clustering model and a test set used only for cluster-label prediction. Covariates are standardized using training-set moments to avoid leakage.

```r
library(MASS)

set.seed(123)
data("Boston", package = "MASS")

dat <- Boston
dat <- subset(dat, select = c(crim, lstat, rm, nox))

n <- nrow(dat)
idx_train <- sample(seq_len(n), size = floor(0.75 * n), replace = FALSE)
train_dat <- dat[idx_train, ]
test_dat  <- dat[-idx_train, ]

# standardize covariates using training set
xvars <- c("lstat", "rm", "nox")
mu_x  <- vapply(train_dat[, xvars], mean, numeric(1))
sd_x  <- vapply(train_dat[, xvars], sd,   numeric(1))

train_dat[, xvars] <- sweep(sweep(train_dat[, xvars], 2, mu_x, "-"), 2, sd_x, "/")
test_dat[, xvars]  <- sweep(sweep(test_dat[, xvars],  2, mu_x, "-"), 2, sd_x, "/")
```

```
  sample   n
1  train 379
2   test 127
```

4.3. Clustering model specification and fit

We fit a bulk-only conditional mixture model for `crim` with three covariates. In this illustration, we select `type="param"`, meaning covariates enter through component-specific regression parameters while the mixture weights are global. This configuration clusters observations by their regression behavior and distributional differences while keeping cluster prevalence interpretable as a global mixing proportion.

The fitted object is subsequently summarized through a representative partition together with cluster-specific response summaries.

```r
fit_clust <- dpmgpd.cluster(
  formula    = crim ~ lstat + rm + nox,
  data       = train_dat,
  kernel     = "lognormal",
  type       = "weights",
  components = 15,
  mcmc       = mcmc_fixed
)
```

4.4. Representative partition and cluster-specific response summaries

Because mixture components are not identifiable up to permutation, the training labels are still extracted from the posterior through the PSM and Dahl's least-squares representative partition. For presentation, however, we focus on the resulting partition and the response distributions within its representative clusters rather than on the full PSM heatmap.

The following command extracts the representative labels for the training sample.

```
z_train <- predict(fit_clust, type = "label")
```

For presentation we summarize the first five most populated clusters from `summary(fit_clust,top_n=5)` and report the associated descriptive statistics in Table 2.

4.5. Cluster labels for new observations

Cluster assignment for new observations is performed through the same prediction interface. The output is a label in the space of the representative training clusters, obtained by aggregating posterior predictive co-clustering evidence at the cluster level.

```
z_test <- predict(
  fit_clust,
  newdata = test_dat,
  type    = "label"
)
```

For the test sample we tabulate only the predicted cluster sizes that are actually represented in the new observations (Table 1).

Table 1: Test cluster sizes

Cluster	C3	C2	C4	C7	C1	C8	C5
n	46	45	23	8	2	2	1

To summarize the dominant representative clusters, Table 2 reports descriptive statistics for up to the five most populated training clusters. The table is displayed with summary metrics in rows and retained clusters in columns, so the cluster sizes, within-cluster means, and within-cluster standard deviations can be compared across clusters directly.

Table 2: Descriptive summaries for up to the five most populated representative clusters

Metric	C3	C4	C8	C2	C7
n	75.000	68.000	47.000	45.000	35.000
crim mean	0.338	0.530	3.843	9.732	7.642
crim sd	1.066	3.464	7.858	16.201	15.235
lstat mean	-0.873	-0.398	-0.136	0.770	0.303
lstat sd	0.373	0.525	0.654	0.823	0.852
rm mean	0.846	-0.159	-0.276	-0.346	0.388
rm sd	0.831	0.564	0.545	0.510	0.864
nox mean	-0.763	-0.815	0.364	0.795	0.563
nox sd	0.463	0.431	0.914	0.802	0.874
certainty mean	0.169	0.144	0.128	0.131	0.147
certainty sd	0.030	0.028	0.028	0.028	0.045

This analysis illustrates the intended clustering workflow: inference is summarized through a representative training partition and its cluster-level descriptive summaries, and new observations are assigned to those clusters using posterior predictive co-clustering evidence. Tables 1 and 2 summarize the predicted test-cluster allocation and the dominant representative training clusters without additional data-analysis plots. In Section 5, we next turn to the causal workflow, which targets treatment contrasts rather than latent partitions.

5. Data analysis II: Causal Inference

This application demonstrates causal estimands from a two-arm spliced DPM regression with a GPD tail. In contrast to Section 4, where the focus is model-based clustering under a CRP bulk-only regression, here the focus is inference on restricted-mean and quantile treatment contrasts, including conditional (covariate-specific) effects and extreme upper-tail behavior.

5.1. Dataset and objective

The AER package does not ship the National Job Training Partnership Act (JTPA) experimental data as an in-package dataset. Instead, we use a bundled causal benchmark data object to provide a clean and reproducible entry point for a heavy-tailed earnings-style application.

Let Y denote post-program earnings (USD), $A \in \{0, 1\}$ denote randomized assignment to training, and $\boldsymbol{x}$ denote baseline covariates including previous earnings and demographics. The scientific goal is to quantify how training changes (i) upper-tail-truncated average earnings and (ii) distributional features of earnings, with special attention to the upper tail.

For this heavy-tailed benchmark, unrestricted mean effects are sensitive to a small number of extreme posterior draws. I therefore use restricted-mean versions of the mean-based causal estimands with cutoff c equal to the empirical 95th percentile of observed earnings. Throughout, we report the four causal estimands in Table 3. Here $Q_a(\tau \mid \boldsymbol{x})$ denotes the arm-specific τ-quantile. The standardized estimands (RMATE, QTE) integrate over the empirical distribution of $\{\boldsymbol{x}_i\}_{i=1}^n$, while the conditional estimands (RMCATE, CQTE) are evaluated at user-chosen covariate profiles.

Table 3: Causal estimands reported in the earnings application

Estimand	Definition	Evaluation
RMATE	$\mathbb{E}[\min\{Y(1), c\}] - \mathbb{E}[\min\{Y(0), c\}]$	Standardized over the empirical covariate distribution
QTE	$Q_{Y(1)}(\tau) - Q_{Y(0)}(\tau)$	Standardized over the empirical covariate distribution
RMCATE	$\mathbb{E}[\min\{Y(1), c\} \mid \boldsymbol{x}] - \mathbb{E}[\min\{Y(0), c\} \mid \boldsymbol{x}]$	Evaluated at representative covariate profiles
CQTE	$Q_1(\tau \mid \boldsymbol{x}) - Q_0(\tau \mid \boldsymbol{x})$	Evaluated at representative covariate profiles

5.2. Spliced DPM regression model

I model arm-specific conditional outcome distributions $F_a(\cdot \mid \boldsymbol{x})$ via a spliced construction:

$$f_a(y \mid \boldsymbol{x}) = \sum_j w_{aj}(\boldsymbol{x})\, k(y \mid \Theta_{aj})\, \mathbf{1}(y \le u) + \{1 - F_a(u \mid \boldsymbol{x})\}\, g(y \mid u, \sigma_a, \xi_a)\, \mathbf{1}(y > u),$$

where the bulk kernel $k(\cdot)$ is a DPM regression (stick-breaking backend) and the tail density $g(\cdot)$ is GPD with scale $\sigma_a > 0$ and shape ξ_a. The GPD tail index ξ_a controls extremal heaviness (e.g., $\xi_a > 0$ corresponds to heavy tails). This model stabilizes inference for extreme quantiles by coupling nonparametric bulk flexibility with EVT-consistent tail extrapolation.

5.3. Model fitting

I fit a two-arm spliced mixture using a lognormal bulk kernel. The MCMC settings below are intentionally modest for vignette speed; in substantive analysis we recommend longer runs and sensitivity checks (e.g., components, threshold priors, and prior scales).

```
fit <- dpmgpd.causal(
  x = y,
  X = X,
  treat = A,
  backend = "sb",
  kernel = "gamma",
  components = 5,
  mcmc = c(mcmc_fixed, list(show_progress = FALSE))
)
```

5.4. Standardized estimands: QTE

I summarize the overall effect using QTE on a quantile grid that extends into the upper tail but stops short of the most unstable 0.99 level in the main text.

```
qte_fit <- qte(fit, probs = taus, show_progress = FALSE)
```

Table 4: Marginal QTE estimates across the main-text quantile grid

index	estimate	lower	upper
0.25	-0.132	-0.293	0.011
0.50	0.639	-0.400	1.465
0.75	1.319	0.713	2.159
0.90	4.129	1.679	7.803
0.95	10.621	-0.851	25.262

Table 4 reports the marginal quantile treatment-effect summaries for the spliced causal model. This table replaces the earlier effect plots in the data-analysis presentation.

5.5. Conditional estimands: restricted mean CATE and CQTE at representative profiles

To illustrate heterogeneity, we evaluate conditional effects at two covariate profiles: (i) a "typical" profile at sample medians and (ii) a "high previous earnings" profile at the 80th percentile of previous earnings, holding other covariates at medians.

```
X_profiles <- if (app2_cache_matches_data(fit, X)) {
  X
} else {
  fit_ps_x <- fit$bundle$design$data$X
  fit_ps_x[, setdiff(colnames(fit_ps_x), "(Intercept)"), drop = FALSE]
}

col_med <- apply(X_profiles, 2, median, na.rm = TRUE)
x_typical <- col_med
x_highprev <- col_med
profile_var <- if ("prevearn" %in% colnames(X_profiles)) {
  "prevearn"
} else {
  colnames(X_profiles)[1]
}
x_highprev[profile_var] <- as.numeric(
  quantile(X_profiles[, profile_var], 0.80, na.rm = TRUE)
)
xnew <- rbind(typical = x_typical, high_prev_earn = x_highprev)
profile_names <- rownames(xnew)
profile_names <- gsub("_", " ", profile_names, fixed = TRUE)
profile_labels <- sprintf("id = %d (%s)", seq_len(nrow(xnew)), profile_names)
```

Table 5: Representative profiles used for conditional causal contrasts; the high-profile shift is applied to x1.

	x1	x2	x3
typical	0.076	-0.096	0.012
high_prev_earn	0.797	-0.096	0.012

```
cate_fit <- cate(
  fit,
  newdata = xnew,
  type = "rmean",
  cutoff = rm_cutoff,
  show_progress = FALSE
)
```

Table 6: Restricted-mean CATE estimates with cutoff c = 8.036

id	estimate	lower	upper
1	0.734	0.213	1.386
2	0.758	0.202	1.323

```
cqte_fit <- cqte(fit, probs = taus, newdata = xnew, show_progress = FALSE)
```

Table 7: Conditional QTE estimates at the representative profiles

	id = 1 (typical)			id = 2 (high prev earn)		
τ	Estimate	Lower	Upper	Estimate	Lower	Upper
0.25	-0.182	-0.267	-0.084	-0.182	-0.267	-0.084
0.50	0.000	0.000	0.000	0.000	0.000	0.000
0.75	0.892	0.551	1.272	0.906	0.561	1.277
0.90	1.101	0.692	1.447	1.380	0.806	1.769
0.95	1.490	0.796	2.253	2.079	1.071	3.094

Tables 6 and 7 summarize the conditional restricted-mean and quantile treatment effects at the representative covariate profiles.

5.6. Tail behavior

A key advantage of the spliced model is direct posterior inference on the tail index ξ_a by arm. Larger ξ_a indicates a heavier upper tail and greater propensity for extreme earnings.

Table 8: Posterior summaries of the arm-specific GPD tail-shape parameters

arm	mean	sd	q0.025	q0.500	q0.975
treated	0.568	0.078	0.483	0.541	0.667
control	0.417	0.139	0.242	0.347	0.667

5.7. Interpretation

The QTE profile shows that the treatment effect strengthens markedly in the upper part of the outcome distribution. The restricted-mean CATE estimates are positive at both representative profiles, while the CQTE curves show stronger upper-tail gains for the higher previous-earnings profile. Finally, the treated arm has the larger posterior tail index ξ_a, indicating a heavier upper tail under training, which is consistent with stronger upper-tail treatment gains in this application.

6. Discussion

CausalMixGPD combines the flexibility of central distribution modeling, the rigor of upper-tail extrapolation, and the ability to compare the full outcome distributions for causal inference in a unified framework. It allows the full flexibility of the non-parametric bulk, the addition of GPD structure in the tails, and the ability to perform a unified inference for the posterior summaries.

The article focuses on the semiparametric aspect of the model, but the software has a wider scope that allows the user to perform a variety of tasks beyond the GPD splicing. It allows the user to perform stick-breaking and Chinese restaurant processes, different kernel families for real-line and positive outcomes, arm-specific causal inference, GPD splicing, and a clustering framework that uses the same machinery as the mixture fitting.

The clustering framework allows the dependence to be modeled through the weights, the atoms, or a combination of both, which is consistent with the dependent DP regression clustering framework. The posterior uncertainty in the clustering partitions is also performed by displaying the similarity matrices and the representative partitions, making the clustering task interpretable even in the presence of label switching in the MCMC samples.

The clustering, bulk tail, and posterior predictive analyses are all connected in a single workflow. This allows for a smooth transition from exploratory subgroup analyses to tail risk assessment and causal inference.

Some Limitations. One limitation of this method is that it is computationally expensive. MCMC for mixture models with covariate-dependent parameters and tail components is computationally expensive, especially when fitting separate chains for each arm, including separate models for each outcome, and including a series of PS estimation models. Larger sample sizes and more dimensions will be even more expensive, especially for sensitivity analyses and model comparison.

Another limitation of this method is that it is a spliced model, and this has all of the tuning parameters of a Dirichlet process mixture and a PoT model. There is a degree of uncertainty in the results, depending on the choice of kernel, truncation, and tail parameters. GPD is available for EVT for high thresholds, but for finite sample tail estimation, there must be sufficient exceedances. Similarly, clustering in a spliced model has similar behavior: strong overlap, weak predictor effects, and/or insufficient exceedances all contribute to a high degree of uncertainty in clustering, where the posterior similarity matrix is more informative than a clustering output.

The causal implementation is currently limited to binary treatments and continuous outcomes, assuming standard identification conditions of consistency, ignorability, and overlap. Propensity score augmentation is helpful for adjustments in observational studies, but it is no replacement for good study design, overlap, and sensitivity analyses for unmeasured confounding. It is not a framework for censored survival, multivariate outcomes, longitudinal treatments, or interference.

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7. Appendix: Additional Theoretical Background

7.1. Generalized Pareto Distribution for Threshold Exceedances

In modeling terms, the tail region is defined by values that exceed a high threshold. Extreme value theory (EVT) gives a standard approximation for exceedances over such a threshold. Let $u(x)$ be a high threshold, which may depend on x, and define the exceedance

$$Z = Y - u(x), \tag{46}$$

considered conditionally on $Y > u(x)$. The conditional excess distribution is given by

$$F_u(z \mid x) = \Pr\big(Y - u(x) \le z \mid Y > u(x), X = x\big), \qquad z \ge 0. \tag{47}$$

Under broad regularity conditions, $F_u(\cdot \mid x)$ is well-approximated by a generalized Pareto distribution (GPD) when $u(x)$ is large (Balkema and de Haan 1974; Pickands 1975; Coles 2001).

For $y > u(x)$, the corresponding GPD cumulative distribution function is given as

$$F_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} 1 - \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0, \end{cases} \tag{48}$$

with support $y \geq u(x)$ when $\xi \geq 0$, and $u(x) \leq y \leq u(x) - \sigma(x)/\xi$ when $\xi < 0$. The corresponding density function is given by

$$f_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} \dfrac{1}{\sigma(x)} \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma(x)} \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0. \end{cases} \tag{49}$$

The associated quantile function is given by

$$Q_{\mathrm{GPD}}(\tau \mid u(x), \sigma(x), \xi) = \begin{cases} u(x) - \sigma(x) \log(1 - \tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\left((1 - \tau)^{-\xi} - 1\right), & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1. \tag{50}$$

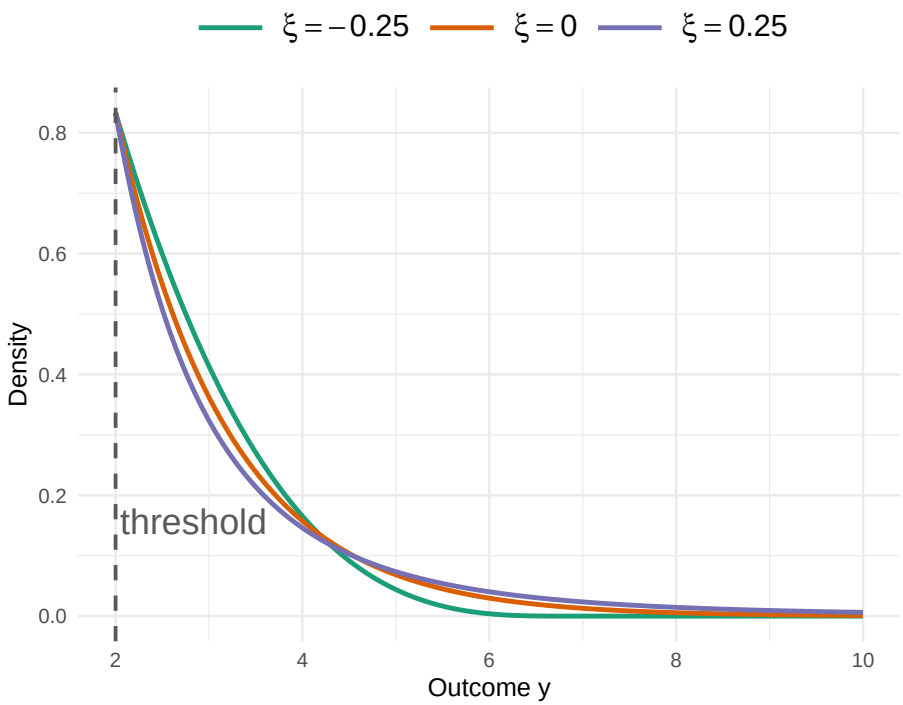

Figure 2: GPD tail density above threshold for multiple shape parameters.

Figure 2, illustrates the GPD density for different shape parameters ξ with a fixed scale parameter σ and threshold u. The shape parameter ξ controls the tail heaviness: $\xi < 0$

corresponds to a light tail with a finite upper bound, $\xi = 0$ corresponds to an exponential tail, and $\xi > 0$ corresponds to a heavy tail with infinite support. The GPD provides a flexible family for modeling exceedances above a high threshold, making it a natural choice for tail modeling in the two part spliced model construction.

Next, we will discuss the modelling of the bulk of the distribution using DPM regression models.

7.2. Dirichlet Process Mixture Model for Conditional Densities

In this package, conditional density of Y given X below a specific threshold ($u(x)$) is modeled using a DPM regression model. Let $k(\cdot \mid x; \theta)$ be a user-chosen kernel family indexed by parameter θ. The hierarchical specification is

$$Y \mid X = x, \theta \sim k(\cdot \mid x; \theta), \qquad \theta \mid H \sim H, \qquad H \mid \kappa, H_0 \sim \mathrm{DP}(\kappa, H_0), \tag{51}$$

where H is an unknown mixing distribution on the kernel-parameter space, H_0 is a base measure, and $\kappa > 0$ is a concentration parameter (Ferguson 1973; Lo 1984). Marginalizing H yields an (almost surely) discrete mixing distribution and an infinite mixture representation:

$$f_{\mathrm{DP}}(y \mid x) = \int k(y \mid x; \theta)\, dH(\theta) = \sum_{j=1}^{\infty} w_j\, k(y \mid x; \theta_j), \qquad w_j \geq 0, \qquad \sum_{j=1}^{\infty} w_j = 1. \tag{52}$$

One representation for the weights is given by Sethuraman's stick-breaking construction (Sethuraman 1994):

$$w_1 = V_1, \qquad w_j = V_j \prod_{\ell<j} (1 - V_\ell), \qquad V_j \overset{\text{iid}}{\sim} \mathrm{Beta}(1, \kappa), \qquad \theta_j \overset{\text{iid}}{\sim} H_0. \tag{53}$$

An equivalent characterization is obtained by marginalizing over H, yielding the Blackwell–MacQueen Pólya urn representation. In particular, the conditional distribution of a parameter value given the others can be written as

$$\theta_i \mid \{\theta_j\}_{j \neq i} \sim \frac{\kappa}{\kappa + n} H_0(\cdot) + \frac{1}{\kappa + n} \sum_{j=1}^{n} \delta_{\theta_j}(\cdot), \tag{54}$$

where $\delta_{\theta_j}(\cdot)$ is a point mass assigned to θ_j . This induces a clustering structure often described via the Chinese restaurant process (CRP) (Blackwell and MacQueen 1973; Neal 2000). Both stick-breaking and CRP representations are widely used for posterior simulation in DPM models (Escobar and West 1995; Neal 2000).

DPM works without specifying the number of mixture components, and can capture complex distributional features such as multimodality, skewness, and heteroscedasticity by choosing an appropriate kernel family. This makes DPMs a natural choice for modeling the bulk of the conditional distribution, where data are abundant and complex features may be present. This package supports multiple kernel families spanning both positive-support and real-line outcomes.

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